Prodigy Anesthesia Quick Review

Valvular Heart Disease

What changes in heart sounds are associated with mitral valve stenosis?

Mitral stenosis is associated with an 'opening snap' that occurs in early diastole and a rumbling diastolic murmur heard best at the axilla or apex.

Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 35.

What is valvular stenosis?

Valvular stenosis is a narrowing of the valvular opening, or orifice, that results in a restriction of flow when the valve is open.

Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 496.

How do the hemodynamic goals for mitral regurgitation differ from those of mitral stenosis? In what ways are they similar?

In mitral regurgitation, your goal is to maintain an increased heart rate and decreased afterload while the goal in mitral stenosis is to maintain a normal or decreased heart rate and normal afterload. In both disorders, you should maintain normal sinus rhythm, avoid increases in pulmonary vascular resistance, and maintain preload at normal to increased levels.

Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 498-500.

What therapies have been shown to be effective in the treatment of symptomatic mitral valve regurgitation?

Angiotensin-converting enzyme inhibitors, beta-blockers, and biventricular pacing have been shown to reduce the regurgitant volume and improve exercise tolerance in patients with symptomatic mitral valve regurgitation.

Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 37-38.

What changes does mitral regurgitation produce in the pulmonary artery occlusion waveform?

A V wave appears on the pulmonary artery occlusion waveform. Although pulmonary artery occlusion pressure can be an unreliable estimate of LVEDV in patients with chronic mitral regurgitation, changes in the V wave amplitude can help estimate changes in the degree of mitral regurgitation.

Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 37-38.

How do heart sounds change with mitral regurgitation?

There is typically a holosystolic murmur at the apex that radiates to the axilla.

Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 37-38.

How does the regurgitant fraction correlate with severity of disease in mitral valve regurgitation?

A regurgitant fraction less than 30% is considered mild, 30-60% is associated with moderately severe symptoms, and a regurgitant fraction above 60% is considered severe.

Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 410-411.

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What two factors determine the pressure gradient across the mitral valve in mitral valve regurgitation?	The systemic vascular resistance and the left atrial pressure. A decrease in SVR or an increase in left atrial pressure will decrease the regurgitant volume. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 410.*
What three factors determine the regurgitant volume in mitral regurgitation?	The size of the mitral valve opening, the heart rate, and the pressure gradient across the valve. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 37.*
What is the primary disturbance caused by mitral valve regurgitation?	Mitral valve regurgitation is associated with a decrease in the forward flow of blood (stroke volume) due to the backward flow of blood into the left atrium during systole. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 410.*
What are the most common causes of acute mitral valve regurgitation?	Acute mitral regurgitation can occur as a result of myocardial infarction, papillary muscle dysfunction, chordae tendinae rupture, trauma to the chest, or infectious endocarditis. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 410.*
What are the primary anesthetic management objectives for the patient with mitral valve stenosis?	Avoid hypervolemia which can precipitate pulmonary edema. Nitrous oxide should be used cautiously as it may increase pulmonary vascular resistance. Drugs that predispose the patient to tachycardia such as pancuronium or ketamine should be used cautiously or avoided altogether. Intraoperative tachycardia can be controlled with opioids (with the exception of meperidine which can cause tachycardia), by increasing the depth of anesthesia or administering a beta-blocker. Phenylephrine is preferred over ephedrine for decreases in blood pressure as it doesn't increase the heart rate. Epidural anesthesia is generally preferred over spinal anesthesia because the decrease in sympathetic activity is not as dramatic. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 409-410.*
What is mitral valve prolapse?	Mitral valve prolapse occurs when one or both mitral valve leaflets dip into the left atrium during systole. It can occur with or without mitral regurgitation. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 38.*

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What valvular abnormalities can be caused by mitral stenosis?	Mitral stenosis results in an increase in right ventricular volume. Severe right ventricular dilation can cause tricuspid and/or pulmonary valve insufficiency. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 408.*
What changes in heart sounds are associated with mitral valve prolapse?	Mitral valve prolapse is associated with a midsystolic click and a late systolic murmur. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 39.*
How can mitral valve stenosis result in pulmonary edema?	The increase in left atrial pressure associated with mitral stenosis is transmitted into the pulmonary vasculature. As pulmonary venous pressure increases above approximately 25 mmHg, fluid can leak into the pulmonary interstitial space resulting in a decrease in pulmonary compliance and increased work of breathing. If the change in pulmonary venous pressure occurs over a long period of time, an increase in pulmonary lymph flow can partially compensate for the fluid accumulation. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 35.*
What is the compensatory mechanism that maintains stroke volume in patients with mild mitral stenosis? What factors may cause this mechanism to fail?	Left atrial pressure increases, which increases the blood flow through the mildly narrowed mitral valve opening. The increase in atrial pressure will fail to maintain stroke volume during tachycardia or atrial fibrillation. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 34-35.*
How are the left atrium, pulmonary vasculature, and right ventricle affected by mitral stenosis?	As flow through the mitral valve opening into the left ventricle is decreased, left atrial pressure increases, resulting in left atrial hypertrophy and distention. The increased pressure is transmitted into the pulmonary vasculature as the volume of pulmonary blood increases. This increased pulmonary vascular pressure represents an increase in right ventricular afterload and will cause right ventricular hypertrophy and failure. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 498.*
How is left ventricular volume affected by mitral stenosis and why?	As the mitral valve orifice narrows, a pressure gradient develops across the valve as a compensatory mechanism to maintain flow through the valve. As the valve opening decreases and the pressure gradient increases, the flow of blood through the opening decreases and left ventricular volume decreases. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 497.*
What cardiac conditions may develop from mitral stenosis?	Congestive heart failure, pulmonary hypertension, and right ventricular failure *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 34-35.*

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What effect does rheumatic heart disease have on the mitral valve? What patients does it affect most? How quickly does it develop?	The incidence of mitral stenosis is higher in females. Over a period of 20-30 years, rheumatic fever causes the mitral valve leaflets to become thickened, the commissure may fuse, and the leaflets and annulus may become calcified. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 34.*
What is the normal mitral valve area?	The normal mitral valve area is 4-6 cm2. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 497.*
What is the most common cause of mitral stenosis?	Rheumatic heart disease *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 34.*
What are the most common symptoms of valvular disorders in general?	The most common symptoms associated with valvular disorders as a whole are congestive heart failure, dysrhythmias, syncope, and angina. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 497.*
What is a mixed valvular disorder?	It is a combination of stenosis and insufficiency, although one disorder is usually considered dominant. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 496.*
What is valvular insufficiency?	Valvular insufficiency is the incomplete closure of a valve which allows blood to regurgitate back into the previous chamber, resulting in inhibited forward flow. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 496.*
What range of mitral valve areas are associated with dyspnea at rest? Dyspnea on moderate exertion? Asymptomatic mitral stenosis?	A mitral valve area of 1 cm2 is associated with a transvalvular gradient of 20 mmHg and dyspnea at rest or with minimal exertion. A mitral valve area between 1 and 1.5 cm2 is associated with dyspnea on moderate exertion. A mitral valve area between 1.5 and 2 cm2 is considered mild and patients are typically asymptomatic. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 409.*
What is the average life span of a patient with aortic stenosis if valve replacement is not performed?	Approximately 75% of patients with severe aortic stenosis will die within three years if aortic valve replacement surgery is not performed. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 41.*

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What are the main anesthetic goals for a patient with aortic regurgitation?	The heart rate should be maintained normal to high, the afterload should be decreased, myocardial depression should be avoided, normal sinus rhythm should be maintained, and preload should be normal to high. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 503.*
What is the typical medical management for a patient with aortic regurgitation?	Afterload reduction with ACE inhibitors is often employed to reduce the transvalvular gradient and facilitate forward flow of blood out of the left ventricle. Digitalis and diuretics are often utilized as well. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 417.*
How does the prognosis of acute aortic regurgitation differ from chronic aortic regurgitation?	In acute aortic regurgitation, the left ventricle does not have adequate time to compensate for the increased left ventricular volume and left ventricular failure, pulmonary edema, and cardiovascular collapse may ensue. Patients with chronic aortic regurgitation may not exhibit symptoms for a long period of time and during normal activity, the symptoms are not incapacitating. As long as the mitral valve patency does not result from left ventricular hypertrophy, the pulmonary circulation is preserved. End-stage disease is typically characterized by myocardial failure, a decrease in cardiac output, dramatic elevations in LVEDV, and pulmonary edema. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 502.*
How does an increase in diastolic blood pressure affect the severity of aortic regurgitation?	An increase in the diastolic blood pressure increases the backward pressure gradient which results in an increase in the proportion of stroke volume that regurgitates back into the left ventricle. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 416.*
How does bradycardia affect the severity of aortic regurgitation?	The decrease in heart rate increases the diastolic time which adversely affects the patient's condition by increasing the proportion of the stroke volume that regurgitates backward through the aortic valve. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 416-417.*
What are the principal hemodynamic effects of aortic regurgitation?	Because of the backward flow of blood through the aortic valve during diastole, the left ventricle becomes volume overloaded and distended. The arterial diastolic pressure and systemic vascular resistance are decreased, which helps maintain forward flow of blood into the arterial tree. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 416-417.*

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What are the most common conditions affecting the aortic valve leaflets that result in aortic regurgitation?	Rheumatic fever, bicuspid aortic valve, infective endocarditis, and the use of anorexigenic (diet) drugs such as fenfluramine or dexfenfluramine. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 42.*
What is aortic regurgitation?	Aortic regurgitation is the failure of the aortic valve leaflets to close properly due to disease of the aortic root or of the leaflets themselves. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 42.*
What are the primary anesthetic management goals for a patient with aortic stenosis?	Because left ventricular filling is so dependent upon an appropriately-timed atrial contraction, maintenance of normal sinus rhythm is very important. The development of a junctional rhythm or atrial fibrillation can result in congestive heart failure and hypotension. Hypotension must be treated aggressively as the corresponding decrease in myocardial perfusion can decrease left ventricular function and result in a decrease in cardiac output which would further worsen hypotension. Sustained increases in heart rate must be avoided as the ventricles need adequate time for ventricular filling. Conversely, severe bradycardia can result in overfilling and ventricular distention. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 41.*
Why is atrial fibrillation or a junctional rhythm so detrimental to a patient with aortic stenosis?	The decreased pressure gradient that exists between the left atrium and left ventricle in aortic stenosis limits left ventricular filling dramatically. Because left ventricular filling is so dependent upon atrial contraction, loss of atrial systole can result in congestive heart failure or hypotension. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 415.*
How is left ventricular compliance affected by aortic stenosis?	Compliance decreases as the left ventricle hypertrophies, resulting in diastolic dysfunction. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 414-415.*
What is the anesthetic of choice in patients with mitral regurgitation?	General anesthesia is the anesthetic of choice. Neuraxial anesthesia is not contraindicated, but the dramatic decreases in sympathetic activity and associated decrease in blood pressure should be strongly considered prior to implementation. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 500.*

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How are myocardial oxygen supply and demand affected by aortic stenosis?	The myocardial demand is increased due to ventricular hypertrophy and the supply is decreased as the extraordinary compression of intramyocardial vessels during systole restricts arterial flow to the myocardium. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 414-415.*
How would the presence of aortic regurgitation affect your selection of muscle relaxants for a patient?	Tracheal intubation can be accomplished with any of the muscle relaxants, but because you wish to maintain a normal to elevated heart rate, pancuronium might be a good choice for its vagolytic properties while the potential for succinylcholine to cause bradycardia should be considered prior to administration. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 503.*
What symptoms are associated with severe aortic stenosis?	The classic triad of symptoms that accompany severe aortic stenosis are angina, syncope, and congestive heart failure. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 501.*
What valve area and transvalvular gradient are associated with severe aortic stenosis?	A valve area less than 0.8 cm2 and a transvalvular gradient greater than 50 mmHg are considered to be severe aortic stenosis. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 40.*
What is the initial compensatory mechanism as aortic stenosis develops?	The initial compensatory mechanism for aortic stenosis is an increase in left ventricular pressure to maintain flow through the narrowed aortic valve. This results in a pressure gradient across the valve. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 40-41.*
What is the normal aortic valve area?	2.5 - 3.5 cm2 *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 40.*
What are the risk factors for the development of aortic stenosis?	The risk factors for aortic stenosis are similar to those for ischemic heart disease (e.g. hypertension and hypercholesterolemia). *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 40.*
What is aortic stenosis and what are the two factors that predispose patients to developing this disorder?	Aortic stenosis is the narrowing of the aortic valve which results in obstruction of blood flow into the aorta. It is the result of degeneration and calcification of the leaflets of the aortic valve or the presence of a bicuspid rather than a normal tricuspid valve. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 40.*

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What factors decrease the amount of prolapse in patients with mitral valve prolapse?	Any factor that maintains a larger ventricular volume will decrease the degree of prolapse. Hypertension, vasoconstriction, drug-induced myocardial depression, and increased preload will decrease the degree of prolapse. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 38-39.*
What factors increase the degree of prolapse in patients with mitral valve prolapse?	As a general rule, the larger the ventricle is, the less prolapse the mitral valve will exhibit (the more the ventricle empties, the greater the amount of prolapse). Factors that increase left ventricular contractility or decrease SVR will result in an increased degree of prolapse. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 39.*
What are the symptoms of mitral valve prolapse?	Although many patients are asymptomatic, patients who do exhibit symptoms may experience chest pain, arrhythmias, mitral regurgitation, infectious endocarditis, embolism, and even sudden cardiac death. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 413.*
What connective tissue disorder exhibits a particularly high incidence of mitral valve prolapse?	Although mitral valve prolapse is more common in patients with any connective tissue disorder, it has a particularly high incidence in patients with Marfan syndrome. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 413.*
What percent of the population has mitral valve prolapse?	Mitral valve prolapse was previously estimated to occur in about 5% of the general population and 15% of women, but newer data demonstrates that the actual incidence is between 1.6% and 2.4% in the U.S. population. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 413.*
How is cardiac output affected by aortic stenosis?	The cardiac output is typically normal at rest but the heart cannot increase output appropriately during periods of exertion, resulting in angina and dyspnea, even in the absence of coronary artery disease. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 414.*

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